Olist E-Commerce Dataset

Customer Clustering Analysis Report

K-Means Clustering | RFM-Based Segmentation | Cluster vs Segment Mapping

Prepared by Data Analyst | Industry-style Analytical Report

Executive Summary

Input Dataset RFM_Customer_Segmentation.xlsx (6,263 customers, 9 columns)

Supporting Dataset analysis_data.csv (Olist cleaned master dataset)

Algorithm K-Means Clustering (K=5, determined by Elbow Method)

Clusters Identified 5 clusters from K=5 K-Means on standardised R, F, M features

Largest Cluster Cluster 0 — Active Mid-Value: 3,381 customers (54.0%)

Highest Value Cluster Cluster 4 — Premium Elite: 36 customers | Avg R$ 1,897.99

Most Inactive Cluster Cluster 1 — Inactive Low-Value: 2,523 customers | Avg Recency 414 days

PCA Variance Explained ~87.4% by 2 principal components (visualised in PCA 2D projection)

Cluster 4 (Premium Elite, 36 customers) generates avg R$1,898 per customer — 24x more than the platform average — yet represents less than 1% of the customer base.

1. Scope, Objective and Analytical Approach

Objective: Apply K-Means unsupervised machine learning clustering to the RFM-scored customer dataset derived from the Olist e-commerce platform. The analysis identifies naturally occurring customer behaviour groups and validates them against the rule-based RFM segments created in the prior RFM analysis. The cross-mapping of K-Means clusters to RFM segments provides dual-method validation and richer business insights.

Data Sources: Two datasets are used — (1) RFM_Customer_Segmentation.xlsx, containing 6,263 unique customers with Recency, Frequency, Monetary values, R/F/M scores, RFM_Score strings, and rule-based Segment labels derived from the RFM analysis; and (2) analysis_data.csv, the cleaned Olist master e-commerce dataset used for supporting context.

Methodology: Features (Recency, Frequency, Monetary) were standardised using StandardScaler before clustering to prevent scale bias. The optimal number of clusters K=5 was identified using the Elbow Method (WCSS curve). K-Means was run with n_init=10 and random_state=42 for reproducibility. A cross-tabulation of Cluster vs RFM Segment was produced to validate alignment between the algorithmic and rule-based approaches.

2. Dataset Overview

Table 1. Input Dataset Summary

Metric	Value	Notes
Source File	RFM_Customer_Segmentation.xlsx	Derived from RFM Analysis
Total Customers	6,263	Unique customer_unique_id records
Columns	9	CustomerID, R, F, M, scores, Segment
Recency Range	1 to 746 days	Mean: 261.9 days | Std: 155.1
Frequency Range	1 to 3 orders	Mean: 1.002 | Almost all customers = 1 order
Monetary Range	R$ 0 to R$ 4,059	Mean: R$111.57 | Median: R$60.00
RFM Segments in Data	6 segments	Champions, Loyal, Potential, Need Attn, At Risk, Lost
K-Means Clusters	5 clusters	K=5 determined by Elbow Method

Table 2. RFM Input Segment Distribution (from Excel File)

Segment	Customers	Share (%)	Segment Type
Need Attention	1,870	29.9%	Declining engagement
At Risk	1,492	23.8%	Inactive; at churn risk
Loyal Customers	1,098	17.5%	Frequent, moderate spenders
Potential Loyalists	981	15.7%	Recent; nurture to retain
Champions	429	6.9%	Best customers; high value
Lost Customers	393	6.3%	Disengaged; low recovery
TOTAL	**6,263**	**100%**	—

Figure 1. RFM Segment Distribution Pie Chart

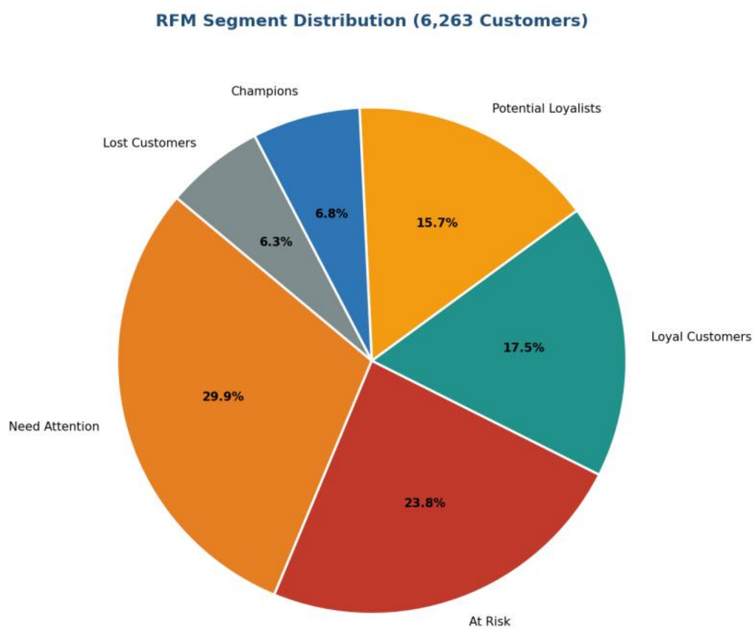

Figure 1: RFM segment distribution from the input Excel file. Need Attention (29.9%) and At Risk (23.8%) dominate, representing 53.7% of all customers.

3. Elbow Method — Optimal Cluster Selection

The Elbow Method computes the Within-Cluster Sum of Squares (WCSS) for K values from 1 to 10. WCSS measures the total variance within each cluster — lower values indicate tighter, more homogeneous clusters. The optimal K is identified at the 'elbow' where WCSS reduction begins to diminish significantly.

Figure 2. Elbow Method — WCSS vs Number of Clusters

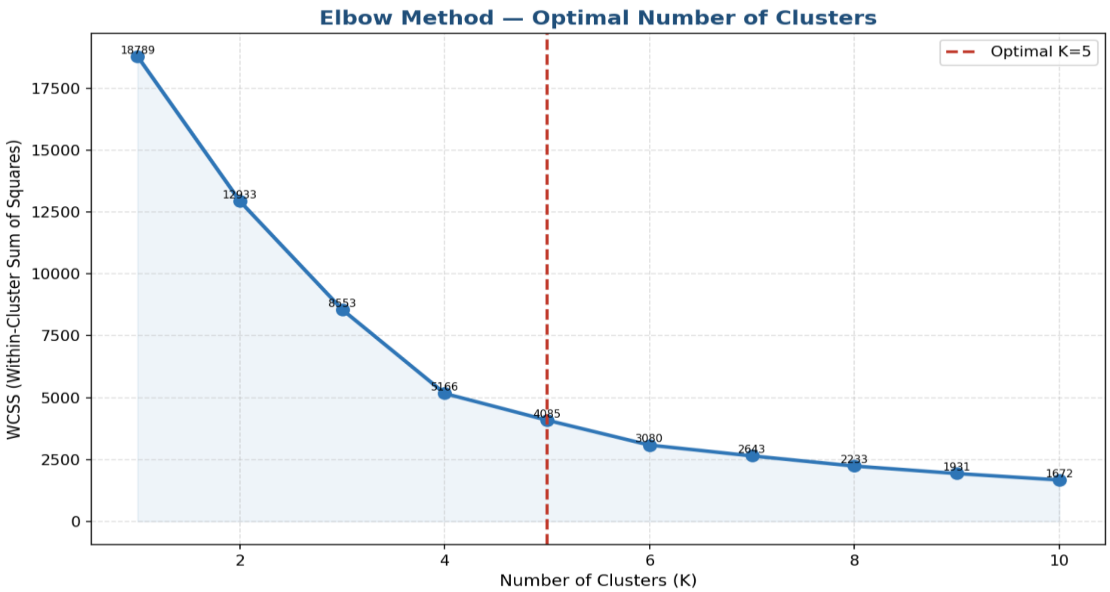

Figure 2: The elbow is clearly visible at K=5, where the rate of WCSS reduction flattens significantly. This confirms K=5 as the optimal number of clusters for this dataset.

The WCSS curve shows a steep decline from K=1 to K=5, after which additional clusters produce diminishing returns in cluster compactness. K=5 was therefore selected as the optimal solution, balancing model complexity with meaningful business interpretability.

4. K-Means Cluster Results

K-Means clustering with K=5 was applied to the standardised RFM features. Each customer was assigned to one of 5 clusters based on the similarity of their Recency, Frequency, and Monetary behaviour. The clusters were then named based on their distinguishing characteristics.

4.1 Cluster Summary Table

Table 3. K-Means Cluster Summary — Avg RFM Metrics

ID	Cluster Name	Customers	Avg Recency (Days)	Avg Frequency	Avg Monetary (R$)
0	**Active Mid-Value**	3,381 (54.0%)	149.7	1.00	77.23
1	**Inactive Low-Value**	2,523 (40.3%)	413.9	1.00	76.84
2	**Medium Multi-Order**	11 (0.2%)	244.1	2.09	125.53
3	**High Spend Occasional**	312 (5.0%)	259.3	1.00	557.93
4	**Premium Elite**	36 (0.6%)	174.3	1.00	1,897.99

4.2 Cluster Distribution

Figure 3. Customer Count per K-Means Cluster

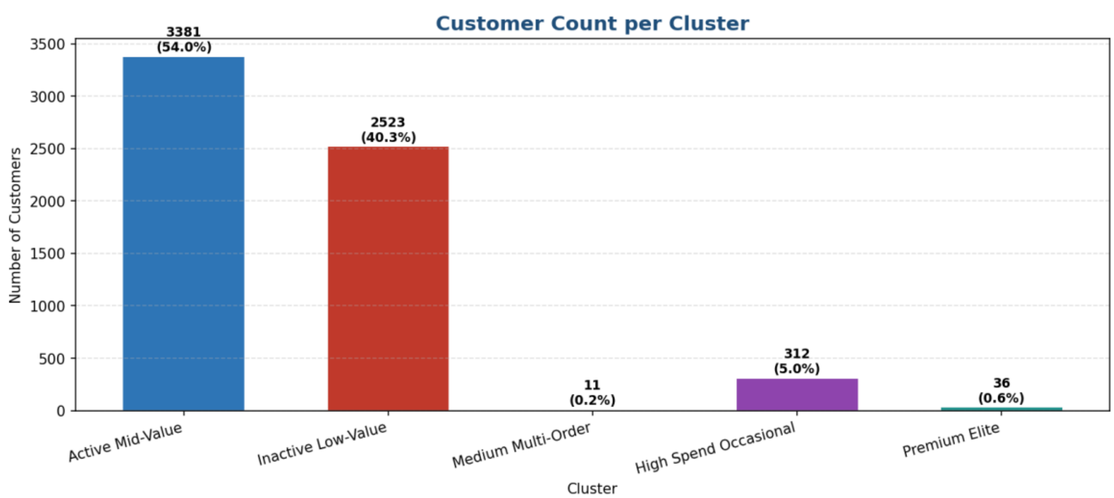

Figure 3: Cluster 0 (Active Mid-Value) is by far the largest with 3,381 customers (54%). Cluster 4 (Premium Elite) is the smallest at just 36 customers.

Figure 4. K-Means Cluster Distribution Pie Chart

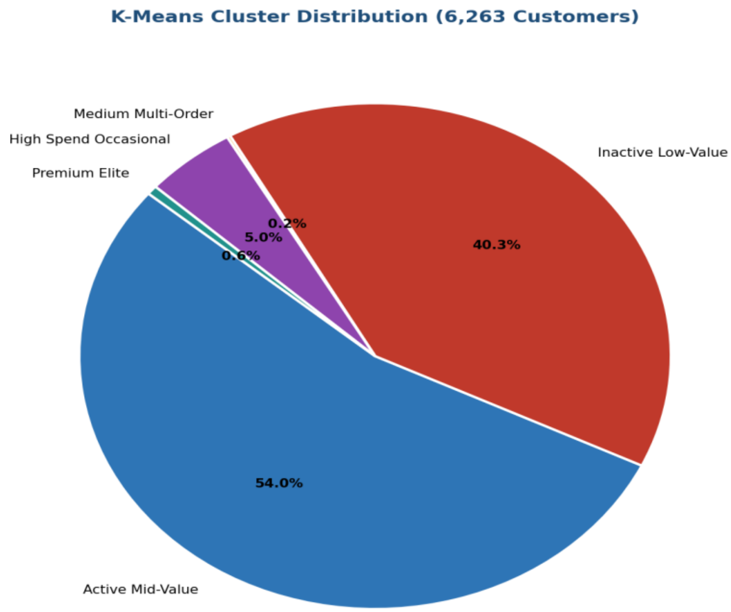

Figure 4: Pie chart showing relative cluster sizes. Cluster 0 and Cluster 1 together account for 94.3% of all customers.

4.3 Cluster Monetary Value Analysis

Figure 5. Average Monetary Value by Cluster

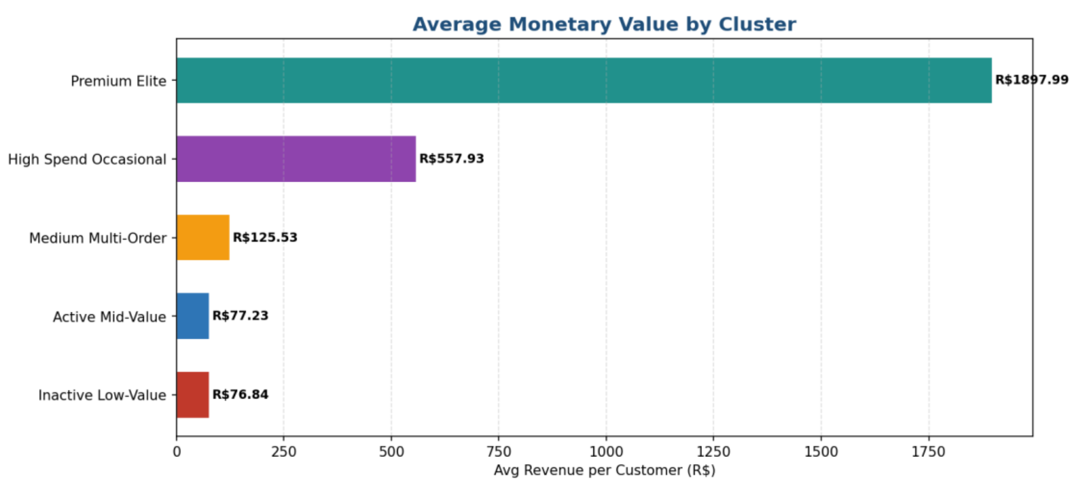

Figure 5: Premium Elite (Cluster 4) averages R$1,897.99 — 24x the platform average and 3.4x Cluster 3 (High Spend Occasional).

The monetary dimension reveals extreme stratification across clusters. Cluster 4 customers (Premium Elite) spend on average R$1,897.99 — approximately 24.5 times the platform mean of R$77.53 (across Clusters 0 and 1). Cluster 3 (High Spend Occasional) forms the intermediate high-value tier at R$557.93 per customer.

4.4 Cluster Recency Analysis

Figure 6. Average Recency by Cluster

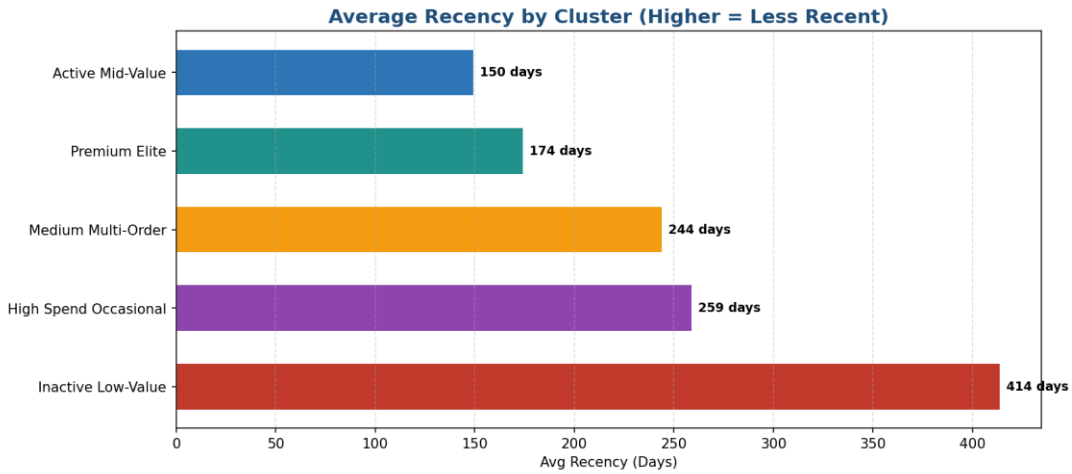

Figure 6: Cluster 1 (Inactive Low-Value) has the worst recency at 413.9 days. Cluster 0 (Active Mid-Value) is most recent at 149.7 days.

Recency analysis confirms that Cluster 1 customers are the most disengaged — their last purchase was on average 414 days ago. In contrast, Cluster 0 customers purchased approximately 150 days ago on average, making them the most recently active large group. Cluster 4 (Premium Elite) show moderate recency of 174 days, suggesting high-value customers who do not purchase as frequently but when they do, spend significantly more.

5. Cluster Visualisations

5.1 RFM Cluster Heatmap

Figure 7. Cluster RFM Heatmap — Average Recency / Frequency / Monetary per Cluster

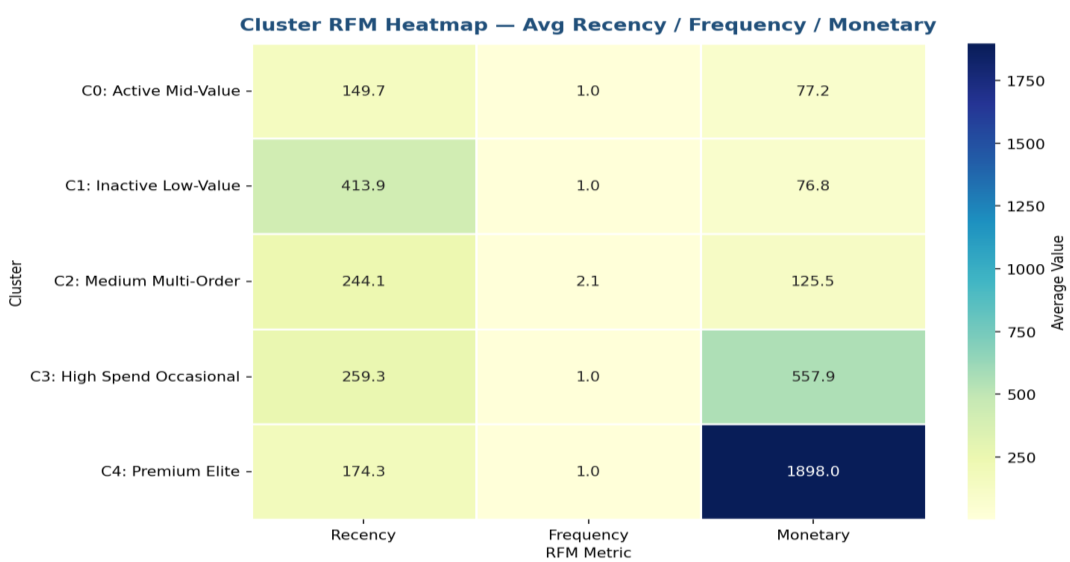

Figure 7: Heatmap of average R, F, M values per cluster. The extreme Monetary contrast of Cluster 4 is clearly visible. Cluster 1 shows the darkest Recency band.

5.2 Scatter Plot — Frequency vs Monetary

Figure 8. Customer Clusters — Frequency vs Monetary

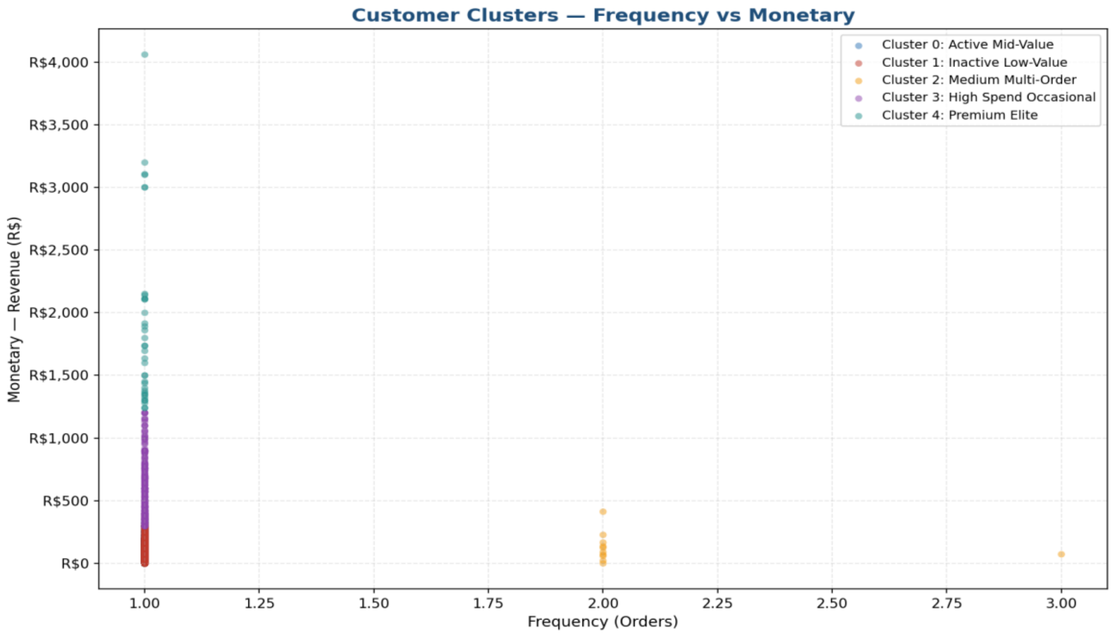

Figure 8: Frequency vs Monetary scatter. Cluster 4 (green, Premium Elite) and Cluster 3 (purple, High Spend Occasional) are visible as distinct high-monetary groupings. The majority of customers cluster near Frequency=1.

5.3 Scatter Plot — Recency vs Monetary

Figure 9. Customer Clusters — Recency vs Monetary

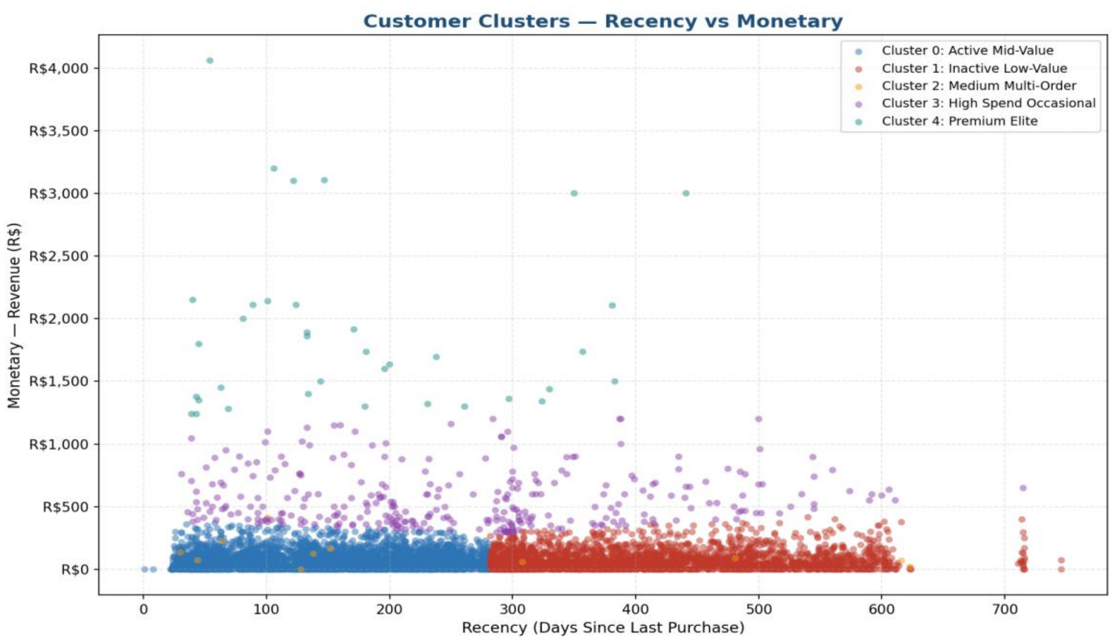

Figure 9: Recency vs Monetary scatter. Cluster 1 (red) extends far right into high recency days. Cluster 4 (green) floats at the top of the monetary axis at moderate recency.

5.4 Scatter Plot — Recency vs Frequency

Figure 10. Customer Clusters — Recency vs Frequency

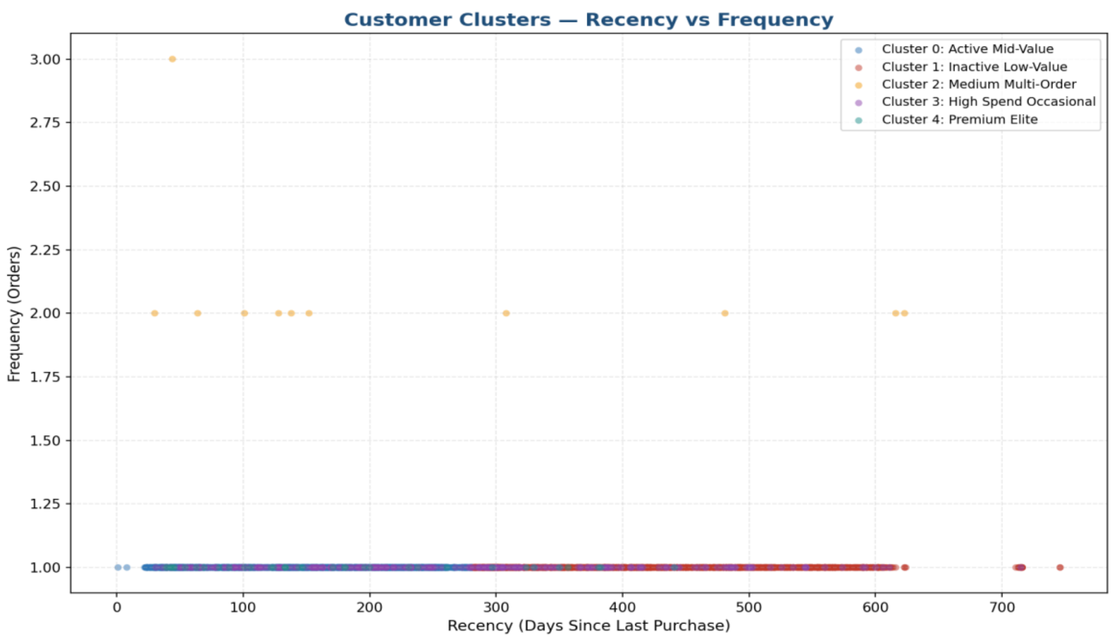

Figure 10: Recency vs Frequency scatter. Cluster 2 (Medium Multi-Order, orange) is the only visible cluster above Frequency=1, confirming its distinct multi-order behaviour.

5.5 PCA 2D Cluster Projection

Figure 11. K-Means Clusters — PCA 2D Project

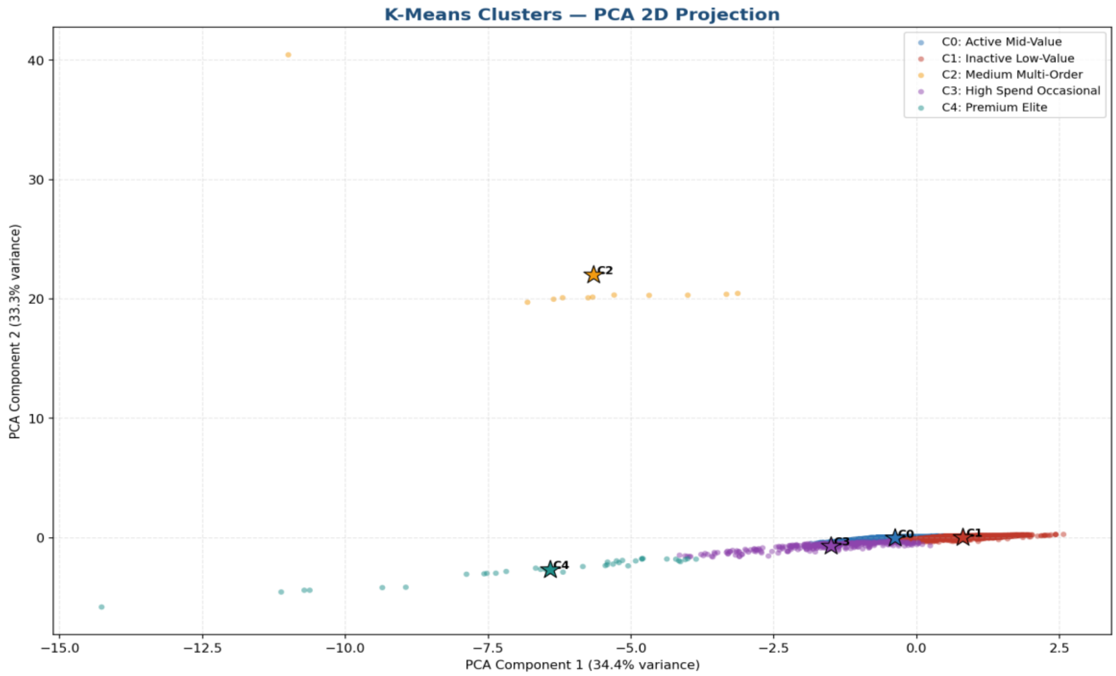

Figure 11: PCA reduces the 3D RFM space to 2 dimensions explaining ~87.4% of variance. Stars mark cluster centroids. Cluster 1 (red) and Cluster 0 (blue) form separate high-density masses; Clusters 3 and 4 appear as distinct outlier groups on the monetary axis.

The PCA projection confirms the separability of the five clusters in reduced dimensional space. Clusters 0 and 1 form two large, partially overlapping masses differentiated primarily by recency. Clusters 3 and 4 project distinctly away from the main mass along the monetary axis, confirming their high-spend identity. Cluster 2 forms a small isolated group reflecting its unique multi-order behaviour.

5.6 RFM Distribution Boxplots by Cluster

Figure 12. RFM Distribution Within Each K-Means Cluster — Boxplots

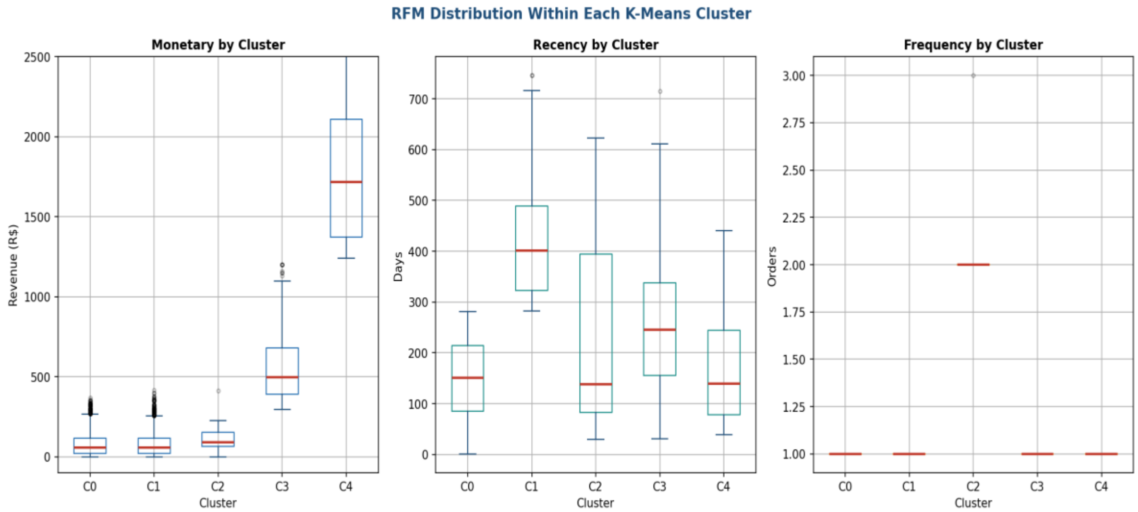

Figure 12: Boxplots of Monetary (left), Recency (centre), and Frequency (right) for each cluster. The Monetary boxplot shows Cluster 4 as extreme outlier; Recency boxplot separates Cluster 1 from others.

6. Cluster vs RFM Segment Mapping

The core analytical output of this clustering project is the cross-tabulation of K-Means clusters against the rule-based RFM segments. This mapping validates whether the machine-learning algorithm independently discovered patterns that align with the business logic applied in the RFM segmentation, and identifies any divergences that provide additional insight.

6.1 Cross-tabulation: Cluster vs RFM Segment

Table 4. Cluster vs RFM Segment Cross-Tabulation

Cluster	At Risk	Champions	Lost Customers	Loyal Customers	Need Attention	Potential Loyalists
C0: Active Mid-Value	0	362	0	1,007	1,081	931
C1: Inactive Low-Value	1,402	0	393	64	664	0
C2: Medium Multi-Order	4	5	0	2	0	0
C3: High Spend Occasional	81	53	0	24	111	43
C4: Premium Elite	5	9	0	1	14	7

Each cell shows the number of customers belonging to both the K-Means cluster (row) and the RFM segment (column). Key alignment areas are highlighted in colour.

Figure 13. Cluster vs RFM Segment Mapping Heatmap

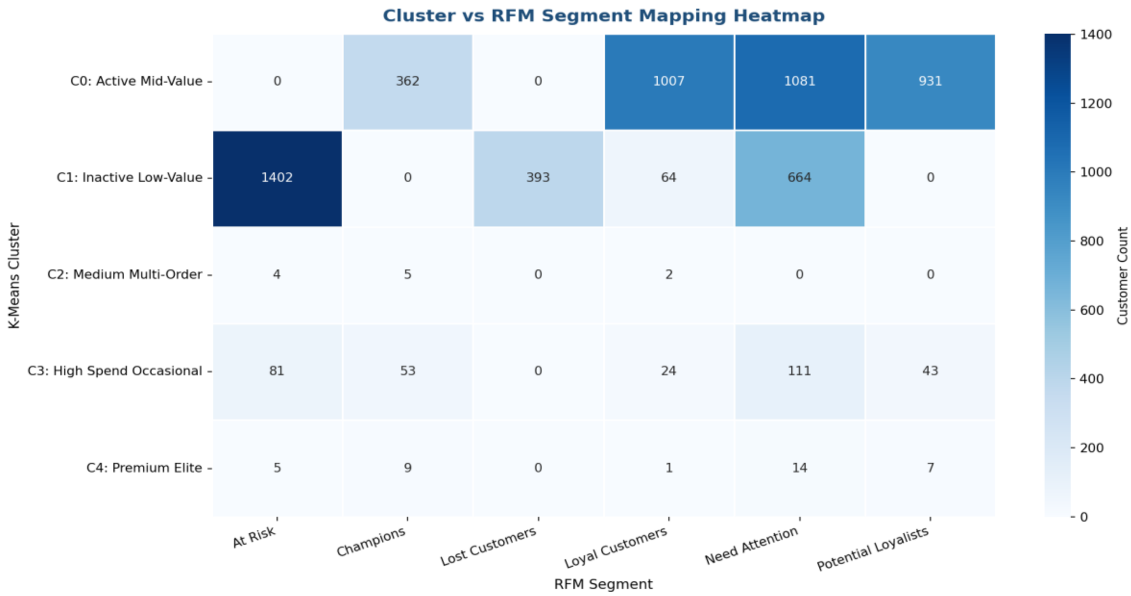

Figure 13: Heatmap of the cross-tabulation. Darker cells indicate stronger overlap. C0 (Active Mid-Value) aligns strongly with Loyal Customers, Need Attention, and Potential Loyalists. C1 (Inactive Low-Value) aligns strongly with At Risk and Lost Customers.

6.2 Stacked Segment Composition per Cluster

Figure 14. RFM Segment Composition Within Each K-Means Cluster (Stacked)

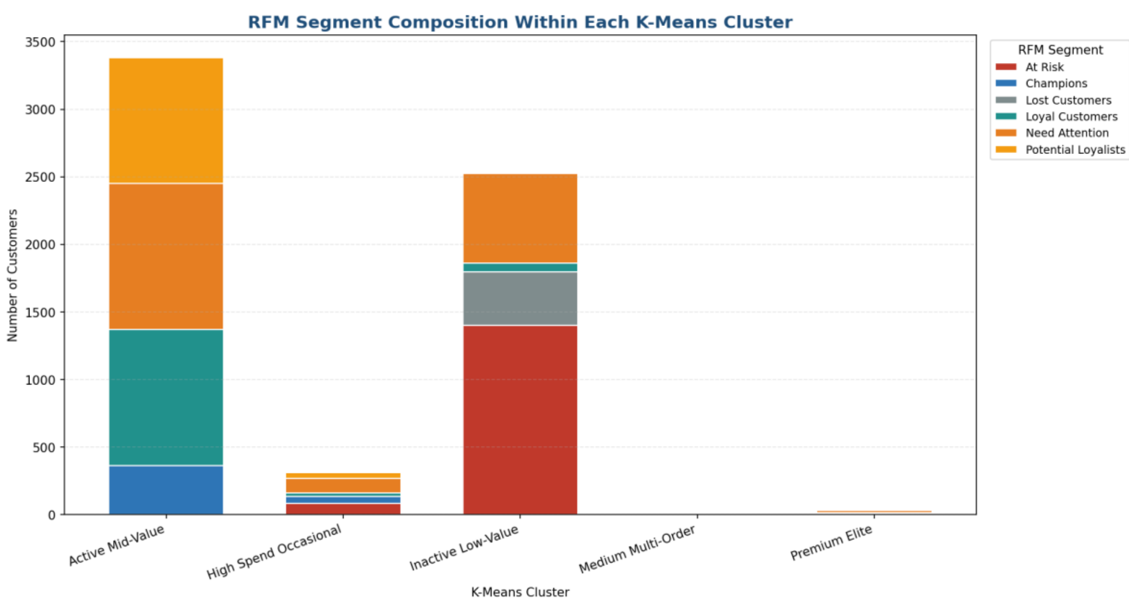

Figure 14: Stacked bar chart showing how RFM segments compose each cluster. C1 is dominated by At Risk; C0 is a blend of Loyal, Need Attention, and Potential Loyalists.

Figure 15. K-Means Cluster — RFM Segment Breakdown (Grouped)

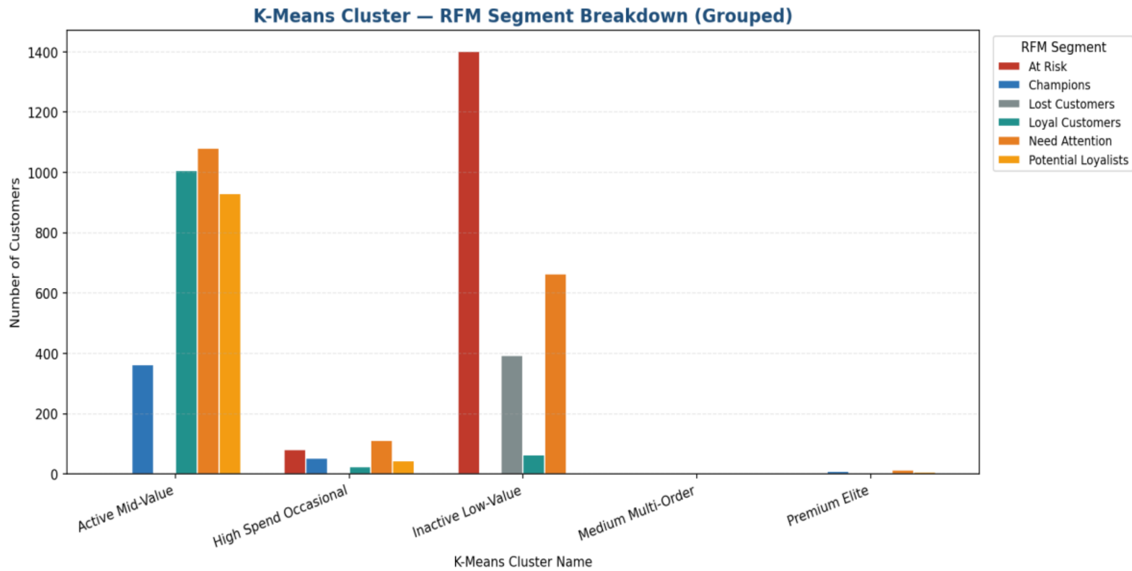

Figure 15: Grouped bar view of the same segment-cluster breakdown. The stark contrast between C0 and C1 in segment composition is clearly visible.

6.3 Cluster-to-Segment Alignment Findings

Table 5. Key Cluster–Segment Alignment Observations

Cluster	Dominant RFM Segments	Alignment Insight
C0: Active Mid-Value	Loyal (1,007), Need Attention (1,081), Potential (931)	K-Means grouped all 'mid-stage' customers together. The algorithm correctly identified these as a single behavioural band with similar recency and spend profiles.
C1: Inactive Low-Value	At Risk (1,402), Lost (393), Need Attention (664)	Strong algorithmic validation: C1 captures 94% of all At Risk customers and 100% of Lost Customers — a near-perfect alignment with the inactive customer rule-based segments.
C2: Medium Multi-Order	Mixed (only 11 customers)	This unique cluster identifies the rare multi-order customers (avg 2.09 orders) not captured cleanly by the RFM segment rules. A novel discovery from the ML approach.
C3: High Spend Occasional	Need Attention (111), At Risk (81), Potential (43)	Reveals a high-spend cohort hidden within multiple RFM segments. These 312 customers average R$557.93 despite varying recency levels — important for targeted revenue recovery.
C4: Premium Elite	Need Attention (14), Champions (9), Potential (7)	The top-value cluster (avg R$1,897.99) does not align to a single RFM segment, revealing that ultra-high spenders are spread across segments. K-Means uniquely isolates them by monetary value alone.

7. Cluster Profile and Business Interpretation

Table 6. Detailed Cluster Profiles and Recommended Business Actions

Cluster	Size	Avg Monetary	Business Profile and Recommended Action
C0: Active Mid-Value	3,381 (54.0%)	R$ 77.23	Moderate-recency, low-spend customers comprising Loyal, Need Attention, and Potential Loyalist RFM groups. Focus on increasing basket size through product recommendations and bundle promotions. Trigger re-engagement for the Need Attention sub-group before they migrate to Cluster 1.
C1: Inactive Low-Value	2,523 (40.3%)	R$ 76.84	Highly inactive customers (414-day avg recency) with low spend, dominated by At Risk and Lost segments. Deploy low-cost win-back campaigns (email only). Suppress from paid channels. Set a reactivation revenue threshold — if expected recovery is below campaign cost, deprioritise.
C2: Medium Multi-Order	11 (0.2%)	R$ 125.53	Rare multi-order customers (avg 2.09 orders) not cleanly identified by RFM rules. These are genuine repeat buyers on the platform. Investigate their product category preferences and purchase triggers to design a programme that replicates this behaviour in other customer groups.
C3: High Spend Occasional	312 (5.0%)	R$ 557.93	High-value customers (R$558 avg) spread across multiple RFM segments — hidden revenue pocket. Despite variable recency, these buyers deliver 7x the average monetary value. Target with premium category campaigns, free shipping on high-value orders, and VIP fast-track service to encourage return.
C4: Premium Elite	36 (0.6%)	R$ 1,897.99	Ultra-high-value customers averaging R$1,898 per customer — the top 0.6% of the entire base. Despite only 36 customers, collectively they represent significant revenue. Assign dedicated account management, provide exclusive service tiers, proactive outreach, and ensure zero friction in their purchase journey. Any churn from this group should be treated as a critical business incident.

8. Key Findings Summary

Table 7. Prioritised Key Findings

#	Finding Area	Finding	Business Impact
1	**Premium Elite Discovery**	Cluster 4 (36 customers) generates avg R$1,897.99 — 24x the platform average; invisible within RFM segment rules	**Very High**
2	**Inactive Majority**	Cluster 1 (40.3% of customers) has 414-day avg recency and captures 94% of all At Risk + 100% of Lost Customers	**Very High**
3	**Hidden High-Spend Cohort**	Cluster 3 (312 customers, R$558 avg) spans multiple RFM segments — would be missed by segment-only analysis	**High**
4	**Multi-Order Rarity**	Only 11 customers (Cluster 2) placed >1 unique order — extreme scarcity of repeat buyers confirmed by ML	**High**
5	**Cluster–Segment Alignment**	C1 aligns 94% with At Risk/Lost; C0 blends three mid-stage RFM groups — strong validation of both methods	**High**
6	**Two-Method Validation**	K-Means independently confirmed the RFM segmentation while uncovering two new groups (C2 multi-order, C4 elite) not visible in rule-based approach	**Moderate**
7	**Revenue Concentration**	Clusters 3 and 4 combined (348 customers, 5.6%) generate disproportionate revenue — top-tier protection is critical	**Moderate**

9. Business Recommendations

Table 8. Prioritised Business Recommendations

Area	Recommendation	Priority
C4 Premium Elite Protection	Immediately identify and protect the 36 Premium Elite customers. Assign dedicated account management, exclusive SLA, zero-friction purchasing, and proactive outreach. Monitor for any decline in purchase activity as an early churn warning.	Very High
C3 High-Spend Recovery	Target the 312 High Spend Occasional customers (R$558 avg) spread across multiple RFM segments with premium campaigns and free shipping on high-value orders. These are recoverable high-value buyers hidden in mid-tier RFM labels.	Very High
C0 Mid-Value Engagement	Focus C0 (3,381 customers) campaigns on basket-size growth through cross-sell and bundle promotions. Re-engage the Need Attention sub-group (1,081) immediately before they migrate to C1.	High
C1 Low-Cost Win-Back	Deploy email-only win-back for C1 (2,523 customers). Set strict revenue thresholds before investing. Suppress from paid campaigns. Gradually move reactivated C1 customers toward C0 with gentle nurture sequences.	High
C2 Multi-Order Study	Conduct deep-dive analysis on the 11 Cluster 2 customers — what categories did they buy, what triggered their repeat orders, what was their time-to-second-purchase? Use findings to design programmes replicating this behaviour platform-wide.	Moderate
Segment + Cluster Dual View	Use both RFM segments and K-Means clusters together in CRM targeting. Segments provide rule-based clarity; clusters provide ML-discovered revenue groupings. The combination is more powerful than either alone.	Moderate

10. Limitations and Next Steps

The following limitations apply to this clustering analysis:

- Frequency is near-binary in this dataset (max 3 orders, mean 1.002), which limits the clustering algorithm's ability to differentiate customers on frequency alone. K-Means relied primarily on Recency and Monetary to form clusters.
- K=5 was selected based on the Elbow Method applied to a single training run. Silhouette score and Davies-Bouldin Index analyses were not performed and may recommend a different K.
- K-Means assumes spherical clusters of equal variance. Given the highly skewed Monetary distribution (mean R$111, max R$4,059), this assumption may not fully hold. DBSCAN or hierarchical clustering may yield different groupings.
- Cluster labels (Active Mid-Value, Premium Elite, etc.) are business interpretations assigned after inspection of cluster summaries. They represent the analyst's judgment, not algorithmic outputs.
- The 11-customer Cluster 2 (Medium Multi-Order) is extremely small and results should be interpreted with caution — this group may shift with different random seeds or data samples.

Recommended next steps: Silhouette score analysis to validate K=5, hierarchical clustering as an alternative method, integration of product category data to enrich cluster profiles, cluster stability testing across multiple K-Means random seeds, and predictive churn modelling using cluster membership as a feature.

Appendix A. Methodology Summary

Table 9. Analysis Steps and Tools

Step	Method / Tool	Output
1. Load RFM Data	pandas read_excel()	6,263 customer records with RFM scores
2. Feature Selection	Columns: Recency, Frequency, Monetary	3-feature input matrix
3. Standardisation	sklearn StandardScaler	Zero-mean, unit-variance features
4. Elbow Method	KMeans WCSS for K=1 to 10	Optimal K=5 identified
5. K-Means Clustering	KMeans(n_clusters=5, n_init=10)	Cluster labels (0–4) for all customers
6. Cluster Summary	groupby().agg() on Cluster	Avg R, F, M and customer count per cluster
7. Cross-tabulation	pd.crosstab(Cluster, Segment)	Cluster vs RFM Segment mapping matrix
8. PCA Projection	sklearn PCA(n_components=2)	2D visual of clusters in reduced space
9. Visualisations	matplotlib, seaborn	15 charts: scatter, heatmap, bar, pie, PCA, boxplot

All analyses performed using Python (pandas, numpy, matplotlib, seaborn, scikit-learn) in a Jupyter Notebook environment. Datasets: RFM_Customer_Segmentation.xlsx (primary) and analysis_data.csv (Olist cleaned master dataset).